

Ahmed Hassan Eldemery

AI Engineer — Machine Learning Engineer

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Professional Summary

AI Engineer and Machine Learning professional building AI agents, computer vision systems, and intelligent automation solutions. Skilled in backend API development, computer vision, and prompt engineering using Python, FastAPI, Scikit-learn, OpenCV, and MongoDB. Passionate about transforming real-world problems into scalable AI products, with hands-on experience delivering production-grade systems spanning multimodal AI classification, SEO automation, and business process automation.

Education

Bachelor of Computer Science (Minor in Artificial Intelligence) Oct 2023 – July 2027 (Expected)

Mansoura University

Mansoura, Egypt

- **Relevant Coursework:** Linear Algebra, Probability & Statistics, Data Structures, Machine Learning, Computer Vision

Technical Skills

- **AI & LLMs:** LLM Integration, Prompt Engineering, LangChain, RAG, Generative AI, Groq API, Llama 3, OpenAI API
- **Machine Learning:** Scikit-learn, XGBoost, Regression, Classification, Clustering, Feature Engineering
- **Computer Vision & Speech:** OpenCV, YOLOv8, PaddleOCR, MediaPipe, TalkNet, Whisper, CLIP
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)
- **Backend & Frontend:** FastAPI, Flask, React, REST APIs, WebSockets, JWT Authentication, PGVector, Qdrant
- **Tools & Architecture:** Python, SQL, Git, Docker, Docker Compose, Microservices, Chrome Extensions (MV3), PostgreSQL, WordPress REST API

Professional Experience

Artificial Intelligence Engineer Jul 2026 – Aug 2026

Howeyah –

Remote

- Contributed to Howeya SEO-AI, an AI-powered SEO platform analyzing, optimizing, and improving WordPress websites through automated AI-driven recommendations and validation workflows.
- Enhanced WordPress integration and automated SEO content and media optimization, including attachment/image metadata handling and Alt Text automation.
- Developed and refined multi-stage validation workflows (live validation, SEO rules, freshness checks) to ensure AI-generated recommendations are accurate, relevant, and safe to apply.
- Improved the platform's crawling engine, data processing, and evidence management capabilities for auditability and traceability.
- Contributed to improving the platform's reliability and production readiness, achieving 21/21 passing PostgreSQL tests and 394/394 passing regression tests.

Artificial Intelligence Specialist

Apr 2026 – Jul 2026

Try GC (Formerly Grand Community)

Mansoura, Egypt – On Site

- Built a multimodal influencer classification system processing **60,000+ profiles**, combining computer vision (face/video analysis) and speech models (TalkNet, Whisper) to auto-classify creators.
- Developed a Manifest V3 Chrome Extension and FastAPI backend to record meetings, transcribe Arabic conversations via Whisper, and extract structured marketing insights using Llama-3.3 via Groq.
- Designed LLM integration pipelines and prompt workflows, reducing manual operational efforts across the company's core database.
- First member of the AI team; helped recruit and onboard university colleagues onto the team.

Machine Learning Engineer Trainee

June 2025 – Dec 2025

Huawei & Mansoura University Collaboration

Mansoura, Egypt

- Engineered end-to-end ML pipelines (Python, Scikit-learn), improving model accuracy by ~12% via feature engineering.
- Trained and evaluated supervised and unsupervised models on real-world datasets.

Machine Learning Engineer Intern

July 2025 – Aug 2025

National Telecommunication Institute (NTI)

Mansoura, Egypt

- Developed a predictive analytics model using Scikit-learn to forecast e-commerce customer spending behavior.
- Executed end-to-end ML data pipelines, enabling data-driven insights from historical sales data.

Key Projects

Howeya SEO-AI (Howeyah)

Python, FastAPI, PostgreSQL, WordPress REST API, Pytest, Docker

- Contributed to an AI-powered SEO platform analyzing, optimizing, and improving WordPress websites through automated AI-driven recommendations and validation workflows.
- Built multi-stage validation pipelines (live validation, SEO rules, freshness checks) and an evidence management system ensuring AI-generated recommendations were accurate, auditable, and safe to apply.
- Improved crawl engine reliability and WordPress media/Alt Text automation, achieving 21/21 passing PostgreSQL tests and 394/394 passing regression tests.

Influencer Video Classifier (Try GC)

Python, OpenCV, TalkNet, Whisper

- Engineered a scalable production pipeline combining computer vision with speech models to classify creators based on content type and speaking style.
- Automated marketing operations by fully replacing manual classification workflows across a massive database.

AI Meeting Assistant (Try GC)

FastAPI, Chrome Extension (MV3), Groq API, Whisper

- Engineered a Manifest V3 Chrome Extension utilizing offscreen documents to bypass service worker constraints for real-time audio mixing and secure microphone capture.
- Built a stateless FastAPI backend with strict Pydantic validation to process audio chunks via Groq's Whisper-large-v3 and extract structured Arabic marketing insights using Llama-3.3 in JSON mode.

VoltIQ — Smart Electricity Platform

FastAPI, Flask, YOLOv8, PaddleOCR, XGBoost, PGVector, Docker

- Architected a microservices AI platform combining computer vision OCR, predictive ML, and a RAG document assistant.
- Built an OpenCV/PaddleOCR pipeline to read LCD 7-segment displays and trained XGBoost models for anomaly detection and balance depletion forecasting.
- Deployed a FastAPI RAG microservice utilizing PGVector/Qdrant and LLMs (Gemini/OpenAI) for context-grounded Q&A.

YAQIZ — AI Workplace Safety & Fatigue Monitoring

FastAPI, React, YOLOv8, MediaPipe, Docker

- Built a full-stack platform combining YOLOv8 PPE detection with ByteTrack worker tracking and MediaPipe fatigue analysis.
- Engineered WebSocket pipelines for live video streaming and containerized the full stack using Docker for scalable deployment.

Digital Employee – AI MSME Sales Assistant

Python, Flask, OpenAI API, MongoDB, JWT

- Built a bilingual LLM chat assistant converting conversations into structured orders with auto-generated IDs.
- Implemented role-based access, low-stock alerts, and config-driven workflows for 10+ business sectors.

ErgoAI – Intelligent Health & Productivity Assistant

Python, OpenCV, Computer Vision

- Developed an AI-powered desktop app utilizing OpenCV to detect blink rate, yawning, and posture at ~20 FPS.
- Achieved **3rd Place** in the Computer Vision Course (Mansoura University) and **3rd Place** at the Ibtikar 8 Innovation Competition.

Data & Academic Projects

Scikit-learn, GMM, Pandas

- Developed predictive E-commerce spending models, USA Adult Income EDA, and unsupervised GMM Image Segmentation pipelines.

Hackathons & Competitions

- **Huawei Developer Competition (Northern Africa):** Advanced to Top 50 teams (Team Sanad) presenting an AI-driven solution in a regional competitive environment (2025).
- **First AI Summit Week Hackathon (Mansoura Univ):** Developed and exhibited the initial prototype of the Digital Employee Assistant alongside teams from 16 universities (2026).
- **Egypt Hackathon 4.0 (GDG Delta):** Built the initial prototype of YAQIZ in a highly competitive hackathon among 120+ teams and 600+ developers (2025).

Honors & Certificates

- **Awards:** 3rd Place – Ibtikar 8 Innovation Comp (2025) — 2nd Place – AI Course Comp (Apr 2025) — Digitopia Semi-Finalist (June 2025).
- **ALX Africa Data Science Program:** 15-month intensive track in Data Analytics and Machine Learning (June 2026).
- **Certifications:** NVIDIA (Building LLM Applications, AI for All) — Intro to Machine Learning (Grade: Excellent) — Applied Data Science & Python.